

1. Wait/Die Scheme

Timestamps

T1=101, T2=102, T3=103, T4=104, T5=105.

T3 R(X)	T3 requests and is granted a read-lock on X
T5 R(X)	T5 requests and is granted a read-lock on X
T1 W(X)	T1 requests an exclusive lock on X, T3 and T5 hold a read-lock on X, T1 is older than both T3 and T5, T1 waits for T3 and T5 to complete
T2 W(Y)	T2 requests and is granted an exclusive lock on Y
T3 R(Y)	T3 requests a read-lock on Y, T2 holds an exclusive lock on Y, T3 is younger than T2, T3 dies/ rolls back and is rescheduled with the original timestamp
T5 W(Z)	T5 requests and is granted an exclusive lock on Z
T5 COMMITS	T5 releases read-lock on X and exclusive lock on Z. T1 is granted exclusive lock on X.
T2 R(X)	T2 requests read-lock on X, T1 holds exclusive lock on X, T2 is younger than T1, T2 dies and is rescheduled with the original timestamp.

2. Why was T1 aborted?

(i) "Uncommitted data" problem.

(ii) Timestamp = 8

(iii) Data that is not committed is a problem because, in this case, Transaction 2 wants access to item C while Transaction 1 is not done with it, i.e. has not committed yet. This means transaction 1 in Time 8 gained access to a temporary data item that was created (Written) in Time 6, and this causes integrity issues whenever there's a

rollback.

The transaction property that is violated is the **isolation property**, which states that data cannot be used by a 2nd transaction until the 1st transaction is done with the data(commits).

3.

FROM vehicle
JOIN constructor **ON** vehicle.conCode = constructor.conCode
JOIN **service ON** vehicle.vin = **service.vin**
WHERE conName = 'Ferrari'
AND serviceType LIKE '%oil%'

Explanation:

The SQL query is now optimised because we are using "JOIN" statements instead of old-style joins. Using JOIN also optimises the filtering process in the WHERE clause. The equality operation is now the first to be performed, which narrows the dataset to only 2 records (since there are 2 Ferraris), and other operations, such as LIKE, can be applied to a smaller dataset, unlike before.